# Antithetic Cross-Validation

PCM Gold Medal Talk

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# 1
# Understanding the situation

# 1 A well known problem

- Suppose we have a sample $Y_{n\times 1} \sim \mathcal{F}_{\mu}$.
- We have one prediction function $g : \mathbb{R}^n \to \mathbb{R}^n$.
- Prediction error: $\text{PE}(g) = \mathbb{E}\left\|g(Y) - \tilde{Y}\right\|^2$.
- Crucial to estimate $\text{PE}(g)$ to get a good predictor $g$.

# Various ways to estimate PE($g$)

# 1.1 A deeper look

For now, we assume $Y_{n\times 1} \sim N(\mu_{n\times 1}, \sigma^2 I_n)$. We will come to a more general case towrds the end of this talk!

$$PE(g) = \mathbb{E}\left\|Y - g(Y)\right\|^2 + 2\sum_{i=1}^{n} \text{Cov}(Y_i, g_i(Y))$$

- The first term can be estimated unbiasedly!
- Only concern is about the second term!

# 1.1 Stein 1981

- For a restricted class (weakly differentiable) of $g$.

$$\text{SURE}(g) = \|Y - g(Y)\|^2 + 2\sigma^2 \sum_{i=1}^{n} \nabla_i g_i(Y)$$

- Unbiasedly estimate $\text{PE}(g)$.

**Definition (Weak Differentiability)**

There exists a function $\nabla g_i : \mathbb{R}^n \to \mathbb{R}^n$, the weak derivative of $g_i$, such that

$$g_i(y+z) - g_i(y) = \int_0^1 z \cdot \nabla g_i(y+tz) \mathrm{d}t,$$

for almost all $y, z \in \mathbb{R}^n$. Denote the Jacobian matrix of $g$ as $\nabla g \in \mathbb{R}^{n \times n}$, where the $i$-th row is equal to $\nabla g_i$.

1. For some constant $\alpha > 0$, generate $Y^{*b}|Y \sim N(Y, \alpha\sigma^2 I_n)$ independently $b = 1, \cdots, B$
2. Estimate the covariance by

$$\widehat{\mathrm{Cov}}_i^{*,\mathrm{b}} = \frac{1}{B-1}\sum_{b=1}^{B}(Y_i^{*b} - Y_i)g_i(Y^{*b})$$

3. Finally estimate the risk by

$$\mathrm{Br}_\alpha(g) = \|Y - g(Y)\|^2 + \frac{2}{\alpha}\sum_{i=1}^{n}\widehat{\mathrm{Cov}}_i^{*,\mathrm{b}}$$

# 1.1 Ye 1998

1. For some constant $\alpha > 0$, generate $Y^{*b}|Y \sim N(Y, \alpha\sigma^2 I_n)$ independently $b = 1, \cdots, B$
2. Estimate the covariance by

$$\widehat{\text{Cov}}_i^{*,y} = \frac{1}{B-1}\sum_{b=1}^{B}(Y_i^{*b} - Y_i)g_i(Y^{*b})$$

3.

$$(s_i^*)^2 = \frac{1}{B-1}\sum_{b=1}^{B}(Y_i^{*b} - \bar{Y}_i^*)^2$$

4. Finally estimate the risk by

$$\text{Ye}_\alpha(g) = \|Y - g(Y)\|^2 + \frac{2}{\alpha}\sum_{i=1}^{n}\frac{\widehat{\text{Cov}}_i^{*,y}}{(s_i^*)^2}$$

# 1.1 Efron 2004

1. For some constant $\alpha > 0$, generate $Y^{*b}|Y \sim N(Y, \alpha\sigma^2 I_n)$ independently $b = 1, \cdots, B$
2. Estimate the covariance by

$$\widehat{\text{Cov}}_i^{*,e} = \frac{1}{B-1}\sum_{b=1}^{B}(Y_i^{*b} - \bar{Y}_i^*)g_i(Y^{*b})$$

3. Finally estimate the risk by

$$\text{Efr}_\alpha(g) = \|Y - g(Y)\|^2 + 2\sum_{i=1}^{n}\widehat{\text{Cov}}_i^{*,e}$$

# 1.1 Problems!

- SURE$(g)$ is defined only for a **restricted class** of $g$!
- $\mathrm{Br}_\alpha(g)$, $\mathrm{Ye}_\alpha(g)$, $\mathrm{Efr}_\alpha(g)$ are **not unbiased**, in fact there is no guideline how to **choose** $\alpha$! (small or big?)

# 1.1 Oliveira et al. 2024, Coupled Bootstrap (CB)

- For $k = 1, \cdots, K$, generate $\omega^k \sim \mathcal{N}(0, \sigma^2 I_n)$ independent of $Y$, and fix $\alpha > 0$;

$$Y^k_{\text{train}} = Y + \sqrt{\alpha}\,\omega^k \quad \text{and} \quad Y^k_{\text{test}} = Y - \frac{1}{\sqrt{\alpha}}\omega^k$$

- 

$$\text{CB}_\alpha = \frac{1}{K}\sum_{k=1}^{K}\left\{\left\|g(Y^k_{\text{train}}) - Y^k_{\text{test}}\right\|_2^2 - \frac{1}{\alpha}\|\omega^k\|_2^2\right\}$$

- For weakly differentiable $g$, as $\alpha \to 0$, $\mathbb{E}(\text{CB}_\alpha | Y)$ recovers $\text{SURE}(g)$!
- But, for fixed $K$,

$$\lim_{\alpha \to 0} \text{MSE}(\text{CB}_\alpha) = \infty!!$$

# 1.1 Liu et al. 2026, Antithetic Cross-Validation (ACV)

- For $k = 1, \cdots, K$, generate $\omega^k \sim \mathcal{N}(0, \sigma^2 I_n)$ independent of $Y$ such that $\text{Cov}(\omega^j, \omega^k) = -\frac{\sigma^2}{K-1} I_n$ and fix $\alpha > 0$;

$$Y^k_{\text{train}} = Y + \sqrt{\alpha}\,\omega^k \quad \text{and} \quad Y^k_{\text{test}} = Y - \frac{1}{\sqrt{\alpha}}\omega^k$$

- Note that, $\sum_{k=1}^{K} \omega^k = 0$ (Zero sum constraint)
-

$$\text{CV}_\alpha = \frac{1}{K}\sum_{k=1}^{K}\left\{\left\|g(Y^k_{\text{train}}) - Y^k_{\text{test}}\right\|_2^2 - \frac{1}{\alpha}\|\omega^k\|_2^2\right\}$$

# 1.1 Properties of ACV

- It also recovers SURE as $\alpha$ goes to 0!

$$\mathbb{E}(\mathsf{CV}_\alpha|Y) = \mathbb{E}(CB_\alpha|Y)$$

- 

$$\lim_{\alpha \to 0} \mathbb{E}(\mathsf{CV}_\alpha) = \mathsf{PE}(g)$$

- Assume $g$ to be weakly differentiable. Then, for fixed $K$,

$$\lim_{\alpha \to 0} MSE(\mathsf{CV}_\alpha) < \infty$$

# 1.1 Breaking the MSE

$$\mathbb{E}\left[\left(\mathrm{CV}_\alpha - \mathrm{PE}(g)\right)^2\right] = \underbrace{\{\mathbb{E}[\mathrm{CV}_\alpha] - \mathrm{PE}(g)\}^2}_{\mathrm{Bias}^2} + \underbrace{\mathrm{Var}\left[\mathbb{E}[\mathrm{CV}_\alpha \mid Y]\right]}_{\text{S-VAR}} + \underbrace{\mathbb{E}\left[\mathrm{Var}[\mathrm{CV}_\alpha \mid Y]\right]}_{\text{R-VAR}}.$$

- $$\lim_{\alpha \to 0} \mathrm{Bias} = 0$$
- **Irreducible Variance:**

$$\lim_{\alpha \to 0} \text{S-VAR} = \mathrm{Var}[\|Y - g(Y)\|^2 + 2\sigma^2 \mathrm{tr}(\nabla g(Y))]$$

- **Reducible Variance:**

$$\lim_{\alpha \to 0} \text{R-VAR} = \frac{4\sigma^4}{K-1}\mathbb{E}[\|\nabla g(Y)\|_F^2 + \mathrm{tr}(\nabla g(Y)^2)]$$

# 2
# A more deeper look into ACV

# Chattopadhyay et al. 2026

**Natural Question**

**Is antithetic randomization is optimal within a broader class of randomization schemes?**

**Our Contributions**

- Antithetic randomization is not only **sufficient but also necessary** for the **limiting MSE** to be **finite** within joint normal CV schemes for weak differentiable $g$.
- Jointly normal antithetic scheme is the **unique minimax-optimal** construction within a **broad class** of antithetic randomization schemes for weak differentiable $g$.
- For **non-smooth estimators** with finitely many **jump discontinuities**, antithetic randomization yields lower MSE than any non-antithetic scheme. Moreover, when the discontinuities are known in closed form, adding a **suitable control variate** restores the **bounded MSE** otherwise available only for smooth estimators.

2.1

SECTION 2 / A MORE DEEPER LOOK INTO ACV

# Necessity of antithetic randomization

# 2.1 Assumptions

**1. Exchangeable normal randomization**

(i) **Exchangeability:** $(\omega^{\pi(1)}, \dots, \omega^{\pi(K)}) \overset{d}{=} (\omega^1, \dots, \omega^K)$ for every permutation $\pi$ of $[K] = \{1, \dots, K\}$.

(ii) **Marginal normality and equicorrelation:** $\omega^k \sim \mathcal{N}(0, \sigma^2 I_n)$ for every $k \in [K]$, and $\mathrm{Cov}(\omega^k, \omega^{k'}) = \rho\sigma^2 I_n$ for every $k \neq k'$, where $\rho \in [-1/(K-1), 1]$.

**2. Weak differentiability and moments at an inflated variance**

All components $g_i$ $(1 \leq i \leq n)$ of $g$ are weakly differentiable. In addition, assume that there exists $\alpha_0 > 0$ such that, for $Z_{\alpha_0} \sim \mathcal{N}(\mu, (1+\alpha_0)\sigma^2 I_n)$, $\mathbb{E}\left\|g(Z_{\alpha_0})\right\|_2^4 < \infty$, $\mathbb{E}\left\|\nabla g(Z_{\alpha_0})\right\|_{\mathrm{F}}^2 < \infty$.

**Theorem (Necessity and sufficiency of antithetic randomization)**

*Suppose that* $(\omega^1, \dots, \omega^K)$ *satisfy 1 for some fixed* $\rho$*, and that* $g$ *satisfies 2 and* $\mathbb{E}\left[\|g(Y) - Y\|_2^2\right] > 0$*. Then, as* $\alpha \downarrow 0$*,*

$$\mathbb{E}\left[\mathrm{Var}(\mathrm{CV}_\alpha \mid Y)\right] = \begin{cases} O(1) & \text{if } \rho = -\frac{1}{K-1}, \\ \Theta\left(\frac{1}{\alpha}\right) & \text{if } \rho > -\frac{1}{K-1}. \end{cases}$$

2.2

SECTION 2 / A MORE DEEPER LOOK INTO ACV

# Minimax optimal antithetic randomization

# 2.2 Broader class

- Antithetic correlation is the **only correlation structure** that keeps the reducible variance bounded.
- Many joint distributions with normal marginals satisfy the **equivalent zero-sum constraint.**
- We therefore ask whether some other antithetic scheme achieves **smaller** reducible variance!

# 2.2 A general construction of antithetic randomization schemes

**General Recipe**

Let $Z \sim \mathcal{N}(0, \sigma^2 I_d)$ with $d \geq n$, and let $M = (M^1, \dots, M^K)$ be an exchangeable family of random matrices in $\mathbb{R}^{n \times d}$, independent of $Z$, satisfying

1. **Co-isometry:** $M^k (M^k)^\top = I_n$ for every $k \in [K]$;
2. **Sum-zero:** $\sum_{k=1}^{K} M^k = 0$.

Define $\omega^k = M^k Z$ for $k \in [K]$. Then the stacked vector $(\omega^1, \dots, \omega^K) \in \mathbb{R}^{nK}$ satisfies Assumption 1 with $\rho = -1/(K-1)$ or, equivalently, the zero-sum constraint $\sum_{k=1}^{K} \omega^k = 0$.

**Jointly normal construction**

Let $Z^1, \ldots, Z^K \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2 I_n)$ with average $\bar{Z} = K^{-1} \sum_{j=1}^{K} Z^j$, and let $Z \in \mathbb{R}^{nK}$ be the stacked vector of $Z^1, \ldots, Z^K$, so that $Z \sim \mathcal{N}(0, \sigma^2 I_{nK})$ and $d = nK$. Let $M^k \in \mathbb{R}^{n \times d}$ be the scaled centering matrix determined by $M^k Z = \{K/(K-1)\}^{1/2}(Z^k - \bar{Z})$.

**Balanced Rademacher; K even**

Let $Z \sim \mathcal{N}(0, \sigma^2 I_n)$, and let $\eta$ be uniformly distributed over the balanced sign vectors $\{\eta \in \{\pm 1\}^K : \sum_{k=1}^{K} \eta_k = 0\}$, independently of $Z$.Set $M^k = \eta_k I_n$, so that $\omega^k = \eta_k Z$.

**Theorem (Minimax optimality of the jointly normal scheme)**

*Let $\mathscr{M}$ denote the class of antithetic schemes constru'cted as per the described procedure, and let $\mathscr{G} = \{g : g \text{ satisfies Assumption 2 with } \mathbb{E}\|\nabla g(Y)\|_F^2 = 1\}$. Then*

$$\inf_{M \in \mathscr{M}} \sup_{g \in \mathscr{G}} \lim_{\alpha \downarrow 0} \mathbb{E}\left[\mathrm{Var}(\mathrm{CV}_\alpha \mid Y)\right] = \frac{8\sigma^4}{K-1},$$

*and $M \in \mathscr{M}$ attains this value if and only if the stacked vector $(\omega^1, \ldots, \omega^K)$ is jointly normal.*

2.3

# Extensions beyond smooth estimators

# 2.3 Form of the functions

$$g(y) = f_0(y) + \sum_{\ell=1}^{m} a_\ell(y)\mathbf{1}\{h_\ell(y) > 0\}.$$

Here, $f_0$ represents the **smooth component** of $g$, while each term $a_\ell(y)\mathbf{1}\{h_\ell(y) > 0\}$ introduces a **jump across the boundary** $\{y : h_\ell(y) = 0\}$.

### 3. Piecewise-smooth predictors

1. **Regularity of the smooth components.** The functions $f_0, a_1, \dots, a_m$ satisfy the regularity conditions in Assumption 2.
2. **Regularity of the thresholds.** For each $\ell = 1, \dots, m$, the function $h_\ell : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz, with Lipschitz constant $L_\ell < \infty$.
3. **Local boundary density and conditional jump moment.** For each $\ell$, there exists $t_0 > 0$ such that $h_\ell(Y)$ has a density $p_\ell$ on $(-t_0, t_0)$. Let $m_\ell(s) = \mathbb{E}\left[\|a_\ell(Y)\|_2^2 \,\middle|\, h_\ell(Y) = s\right]$. Assume that

$$\operatorname*{ess\,sup}_{|s| \le t_0} \; p_\ell(s) m_\ell(s) < \infty.$$

**Theorem (Reducible variance for non-smooth predictions)**

*Suppose that* $(\omega^1, \dots, \omega^K)$ *satisfy Assumption 1, that* $g$ *satisfies Assumption 3, and that* $\mathbb{E}[\|g(Y) - Y\|_2^2] > 0$. *Then, as* $\alpha \downarrow 0$,

$$\mathbb{E}\left[\mathrm{Var}\{\mathrm{CV}_\alpha \mid Y\}\right] = \begin{cases} O(\alpha^{-1/2}) & \text{if } \rho = -\frac{1}{K-1}, \\ \Theta(\alpha^{-1}) & \text{if } \rho > -\frac{1}{K-1}. \end{cases}$$

2.4

# Introducing the Control Variate

For each $\ell \in [m]$, let

$$r_\ell(\omega^k, Y) = \mathbb{E}\Big[\omega^k \mathbf{1}\{h_\ell(Y + \sqrt{\alpha}\omega^k) > 0\} \,\Big|\, Y\Big] - \omega^k \mathbf{1}\{h_\ell(Y + \sqrt{\alpha}\omega^k) > 0\}.$$

Define the control variate

$$\mathscr{V}_\alpha = \frac{2}{K\sqrt{\alpha}} \sum_{k=1}^{K} \sum_{\ell=1}^{m} \Big\langle a_\ell(Y), r_\ell(\omega^k, Y) \Big\rangle,$$

and the adjusted estimator

$$\widetilde{\mathsf{CV}}_\alpha = \mathsf{CV}_\alpha + \mathscr{V}_\alpha.$$

# 2.4 Properties of $\widetilde{\mathrm{CV}}_\alpha$

- $$\mathbb{E}(\mathscr{V}_\alpha \mid Y) = 0$$

- Hence, $\widetilde{\mathrm{CV}}_\alpha$ and $\mathrm{CV}_\alpha$ has same conditional mean, and therefore the same **bias** and **S-VAR**!
- The adjustment affects only the **reducible variance**!

**Theorem (Bounded reducible variance with control variates)**

*Let* $(\omega^1, \dots, \omega^K)$ *satisfy Assumption 1 for some fixed* $\rho$*. Let* $g$ *satisfy the smooth-component conditions in Assumption 3, and assume that* $\mathbb{E}\left[\|f_0(Y) - Y\|_2^2\right] > 0$*. Then*

$$\mathbb{E}\left[\mathrm{Var}\left(\widetilde{\mathrm{CV}}_\alpha \mid Y\right)\right] = O(1) \quad \text{as } \alpha \downarrow 0 \quad \text{if and only if} \quad \rho = -1/(K-1).$$

# 3
# An Example

# 3 A non-smooth ridge regression estimator with hard thresholding

- Let $X \in \mathbb{R}^{n \times p}$ be the fixed design matrix
- Fix a regularization parameter $\lambda > 0$.
- **Ridge Estimator:**

$$\hat{\beta}_{\lambda}^{\text{ridge}}(y) = (X^{\top}X + \lambda I_p)^{-1}X^{\top}y.$$

- **Hard Thresholded Ridge Estimator:**

$$\hat{\beta}_{\lambda,\tau_j,j}^{\text{HT}}(y) = \hat{\beta}_{\lambda,j}^{\text{ridge}}(y) \cdot \mathbf{1}\left\{\left|\hat{\beta}_{\lambda,j}^{\text{ridge}}(y)\right| > \tau_j\right\}, \qquad j = 1, \dots, p,$$

$$Y_{n\times 1} = X_{n\times p}\beta_{p\times 1} + \varepsilon_{n\times 1}, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$$

1. $n = 200, p = 50, s = 10$ nonzero coefficients drawn from Unif$(-1, 1)$.
2. $X$ has independent standard normal entries. $\sigma^2 = \text{Var}(X\beta)/2$
3. $K = 6, \lambda = 10, \tau_j = 1.65 \times \text{sd}(\hat{\beta}^{\text{ridge}}_{\lambda,j}(y))$
4. Each simulation is repeated 100 times independently.
5. Codes are available in here.

# 3 Plots

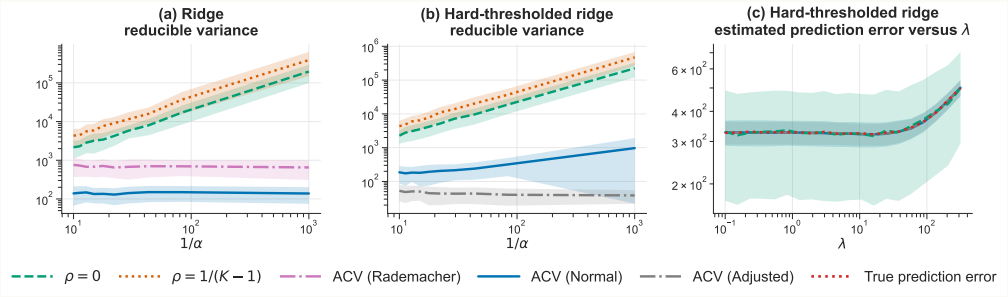



Figure 1: Reducible variance of $\mathrm{CV}_\alpha$ against $1/\alpha$ for ridge regression, panel (a), and for coordinatewise hard-thresholded ridge regression, panel (b); estimated and true prediction error of the hard-thresholded ridge estimator against $\lambda$, panel (c).

# 3 Plots

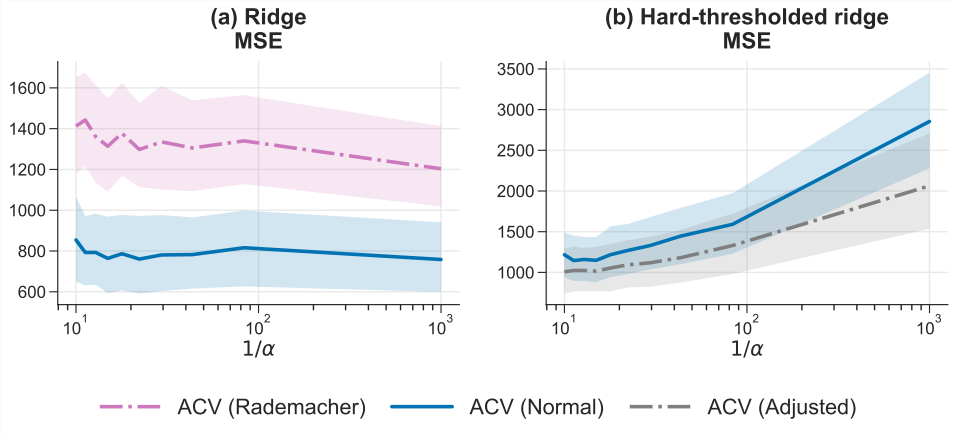



Figure 2: Cross-validation for ridge estimators with and without thresholding. Panel (a) plots the MSE of cross-validation estimators for ridge regression against $1/\alpha$. Panel (b) plots the MSE for coordinatewise hard-thresholded ridge regression against $1/\alpha$.

# 4

# Extension to Exponential Family

# 4 Extensions

- Suppose $Y_n \sim \exp\{\sqrt{n}(\theta_n^T S_n(Y_n) - A_n(\theta_n))\} \cdot h_n(Y_n)$.
- Assume

$$H_n^{-1/2}(S_n - \mu_n) \xrightarrow{\mathscr{L}} N(0, I_p)$$

- Then, we can perform all the details with the rescaled sufficient statistics $T_n = H_n^{-1/2} S_n$
- All the results will hold asymptotically ($n \to \infty, \alpha \to 0$).

# 5
# References

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# Thank you

Questions are welcome!

Paper URL



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